

FOOD DELIVERY SYSTEM

A MINI PROJECT REPORT

submitted by

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BONAFIDE CERTIFICATE

Certified that this project report “ FOOD DELIVERY SYSTEM” is the bonafide work of “KAVYASRI.V(240701249), MAHALAKSHMI.A (240701300),KIRUTHIKA.S(240701265)” Who carried out the project work Submitted for the practical exam

ABSTRACT:

- A food delivery system is the process used to transport prepared food from restaurants,
- kitchens or food production centers to customers. This system has developed rapidly in the
- past few .Due to the habits in the lockdowns and restrictions from the COVID-19
- pandemic, online food delivery through third-party companies has become a growing
- industry and caused a delivery revolution. Decades due to technological developments,
- shifting consumer preferences, and the growth of online apps

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

food delivery system refers to a service or infrastructure that facilitates the transportation of food from a provider, such as a restaurant, grocery store, or meal preparation service, to a consumer's location. It has become increasingly popular due to its convenience, speed, and accessibility, driven by technological advancements and changes in consumer preference

1.2 OBJECTIVES

- The primary objective of a food delivery system is to ensure timely and efficient delivery of food while maintaining its quality and safety. It aims to provide a seamless customer experience by offering user-friendly platforms for ordering and real-time tracking of deliveries. The system strives to optimize resources and minimize costs, ensuring scalability to handle increased demand without compromising service quality. Sustainability is also a key focus, with efforts to use eco-friendly packaging and reduce environmental impact. Additionally, customer satisfaction is prioritized by integrating feedback to continuously improve services, making the food delivery process reliable, accessible, and efficient.

1.3 MODULES

1.RESTURANT TO CUSTOMER Restaurant-to-customer delivery which is the entire process of preparing, packaging and transporting the meals from a restaurant to the customer's desired location. This starts when a customer places an order through a delivery app, online or in person, with a restaurant. Restaurant-toConsumer delivery segment includes the delivery of meals carried out directly by restaurants. The order may be made through platforms or directly through a restaurant website or app. These platforms collect the menus of independent restaurants and specialized delivery services are fully packed in a way that preserves freshness and avoids liquid spill in the cover after it is prepared

2. PETTY FOODS

- operated by an individual or a small team, usually with minimal
- initial investment. These businesses are typically informal or
- home-based and cater to local markets. These businesses are
- one of the growing sector in the society, delivery from these
- small sources and small workshops are available Examples A
- petty food business refers to a small-scale food venture, often
- include: Selling homemade snacks, baked foods, or meals etc
-

3. GROCERY

- Grocery delivery service is a convenient service where one can order
- groceries online and simply sit back and relax while items will find way to
- their doorstep. It saves time and labour-work, in-person shopping as well
- as the effort of visiting the local store, which can be particularly
- advantageous for people with busy schedules and mobility difficulties, or
- simply those who prefer not to visit crowded markets. Grocery delivery
- services are generally consist of the food service systems, providing fresh
- produce, dry goods, and cleansing things with same-day, and planned
- delivery. These platforms also offer the delivery of fresh veggies and fruits
- from the nearby supermarket at the same time it can also deliver packed
- foods such as bread,chips, carbonated drinks which helps the humans
- from not going to the crowded supermarket

ADMIN SLIDE MODULES

- ***Admin Module*** The Admin module serves as the backbone of the food delivery system, enabling efficient management and oversight.
 1. ***Dashboard:*** Provides an overview of orders, deliveries, revenue, and customer activity.
 2. ***Order Management:*** Allows tracking and managing all incoming orders, cancellations, and special requests.
 3. ***Restaurant and Menu Management:*** Generates reports on customer behavior, order menus, including pricing.
 4. ***Delivery Personnel Management:*** Manages the assignment, tracking, and performance of delivery staff.
 5. ***Payment and Finance Tracking:*** Monitors payment transactions, refunds, commissions, and overall revenue.
 6. ***Analytics and Reports:*** trends, and operational efficiency.
 7. ***User Management:*** feedback, and complaint resolutions. Handles customer profiles,
 8. ***Promotions and Offers:*** and management of discounts, coupons, and loyalty program. Facilitates the creation

CUSTOMER SLIDE MODULES

- ***Customer Module*** The Customer module focuses on delivering a seamless and user-friendly experience for users. 1. ***User Registration and Login:*** Allows customers to create profiles, manage accounts, and save preferences. 2. ***Restaurant Browsing:*** Enables searching for restaurants by location, cuisine, rating, and offers. 3. ***Menu Selection and Ordering:*** Provides a smooth interface for browsing menus, customizing items, and placing orders. 4. ***Order Tracking:*** Offers real-time updates on order preparation and delivery status. 5. ***Payment Integration:*** Supports multiple payment options like credit/debit cards, digital wallets, and cash on delivery. 6. ***Notifications:*** Sends alerts for order confirmation, status updates, offers, and promotions. 7. ***Feedback and Support:*** Allows users to rate orders, give feedback, and raise complaints or queries. 8. ***Order History:*** Displays past orders for quick

CHAPTER 2 SURVEY OF TECHNOLOGIES

2.1 SOFTWARE DESCRIPTION

The food delivery System is designed with a clear separation between the customer and owner, ensuring an efficient and organized structure for user interactions and data management. The system uses c language for building the user interface,enabling users to navigate the system with ease. SQL is used as the backend database management system to store, retrieve, and manipulate data securely and effi

2.2 LANGUAGES

2.2.1 C PROGRAMMING LANGUAGE

C programming is a powerful, general-purpose, and procedural programming language widely used for system-level and application development. Known for its simplicity and efficiency, C provides low-level access to memory and a rich set of built-in functions, making it ideal for developing operating systems, embedded systems, and software requiring high performance. Its modular approach, using functions and structures, enables code reusability and scalability. Though it lacks advanced features like object-oriented programming, C remains a foundational language, forming the basis of many modern programming languages and serving as an excellent tool for understanding core programming concepts.

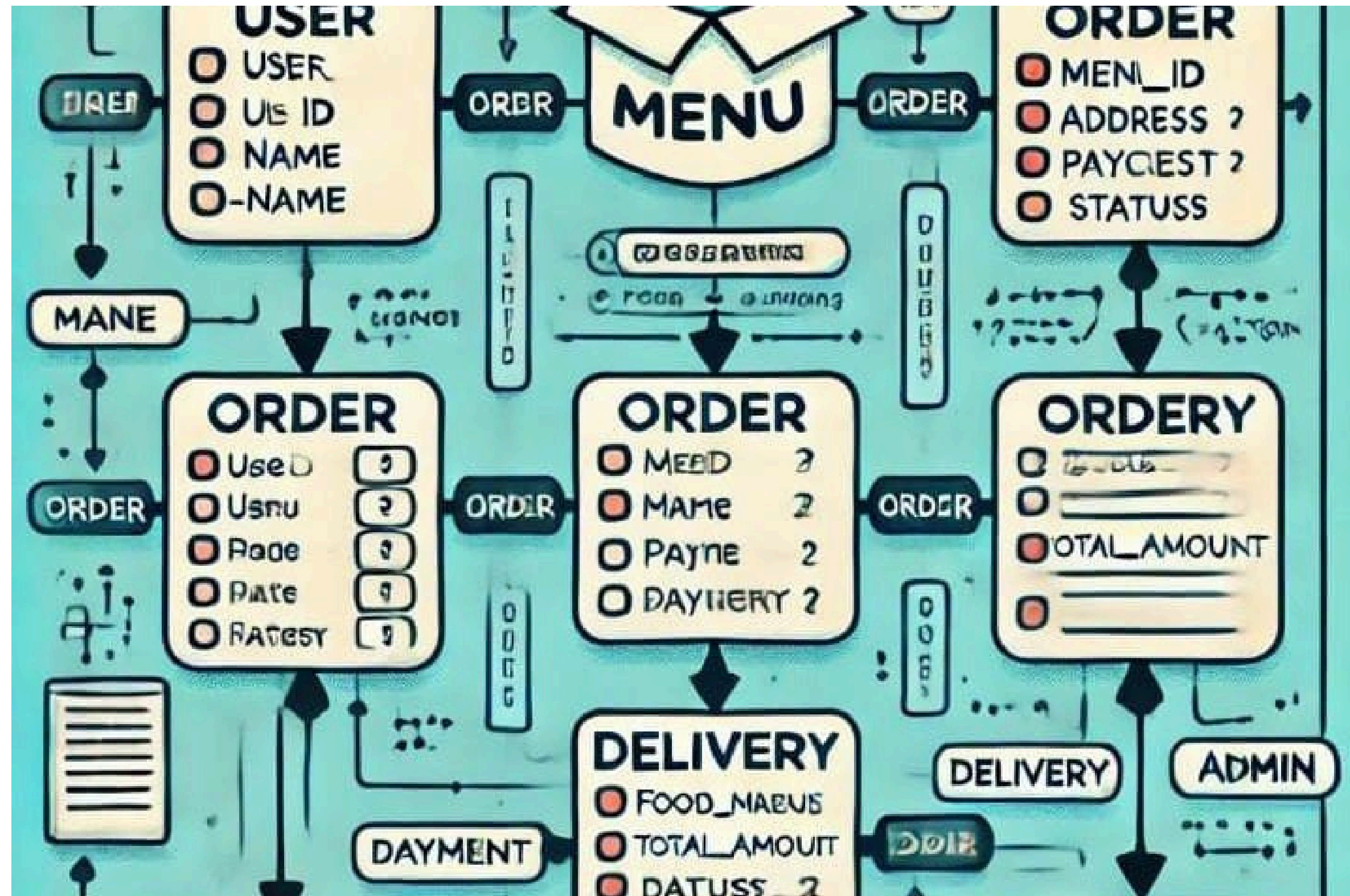
CHAPTER 3 REQUIREMENTS AND ANALYSIS

- 3.1 FUNCTIONAL REQUIREMENTS
- These specify what the system should do. The key functions of a food delivery system may include:
 1. *User Registration & Authentication:* - Users can create accounts by providing necessary details (name, email, phone number, etc.). - Users can log in to their account securely using a username and password.
 2. *Menu Display:* - The system should display the list of available food items, categorized into various sections (e.g., appetizers, main course, drinks, desserts). - Each food item should display its name, description, price, and available quantity.
 3. *Order Placement:* - Users can select food items from the menu and add them to their cart. - Users can modify their cart by adding/removing items before checkout. - The system should calculate the total price, including applicable taxes and discounts.
 4. *Order Confirmation:* - Users should receive a confirmation of their order with details (item names, quantity, price, and estimated delivery time).
 5. *Payment Processing:* - Users can pay for their order using different methods (e.g., credit/debit card, digital wallets, cash on delivery). - The system should process the payment securely and generate a payment receipt.
 6. *Order Tracking:* - Users should be able to track the status of their order in real-time (e.g., placed, preparing, out for delivery, delivered).
 7. *Delivery Management:* - The system should allow users to input their delivery address. - Delivery personnel can update the status of delivery, and users can view real-time updates.

3.2 NON FUNCTIONAL REQUIREMENTS

- These specify how the system should perform and the qualities it should possess.
 1. *Performance:* - The system should be responsive with minimal lag time when users place orders or view the menu. - Order processing time should be fast (e.g., placing orders, updating cart, checking out).
 2. *Scalability:* - The system should be able to handle an increasing number of users and orders without performance degradation.
 3. *Security:* - Sensitive user information (e.g., passwords, payment details) should be encrypted. - The system should be resistant to common security threats like SQL injection, cross-site scripting, and unauthorized access.
 4. *Availability:* - The system should be available 24/7 without downtime for most users, allowing them to place orders at any time.
 5. *Usability:* - The system should have a user-friendly interface with clear instructions for users to navigate and place orders easily. - The menu should be easy to browse, and the checkout process should be straightforward.
 6. *Reliability:* - The system should function correctly under normal usage conditions and handle errors gracefully. - The system should log errors and notify administrators when there's a failure or issue.
 7. *Maintainability:* - The code should be modular, with well-commented functions, so future modifications

3.3 ER DIAGRAM



CHAPTER 4 PROGRAM CODE

- CODE FOR FOOD DELIVERY SYSTEM

```
Data Structures
import time
class Restaurant:
    def __init__(self, name, menu):
        self.name = name
        self.menu = menu
class MenuItem:
    def __init__(self, name, price):
        self.name = name
        self.price = price
class Order:
    def __init__(self, customer_name, restaurant, items):
        self.customer_name = customer_name
        self.restaurant = restaurant
        self.items = items
        self.order_time = datetime.now()
        self.total_price = sum(item.price for item in items)
```

```
# Food Delivery System class
class FoodDeliverySystem:
    def __init__(self):
        self.restaurants = []
        self.orders = []
    def add_restaurant(self, restaurant):
        self.restaurants.append(restaurant)
    def list_restaurants(self):
        return [restaurant.name for restaurant in
self.restaurants]
    def get_menu(self, restaurant_name):
        for restaurant in self.restaurants:
            if restaurant.name == restaurant_name:
                return restaurant.menu
```

```
def place_order(self, customer_name, restaurant_name, item_names):
    restaurant = next((r for r in self.restaurants if r.restaurant_name), None)

    if not restaurant:
        raise ValueError("Restaurant not found.")
    items = [item for item in restaurant.menu if item.name in
item_names]
    if= Order(customer_name, restaurant,
items)
    self.orders.append(order)

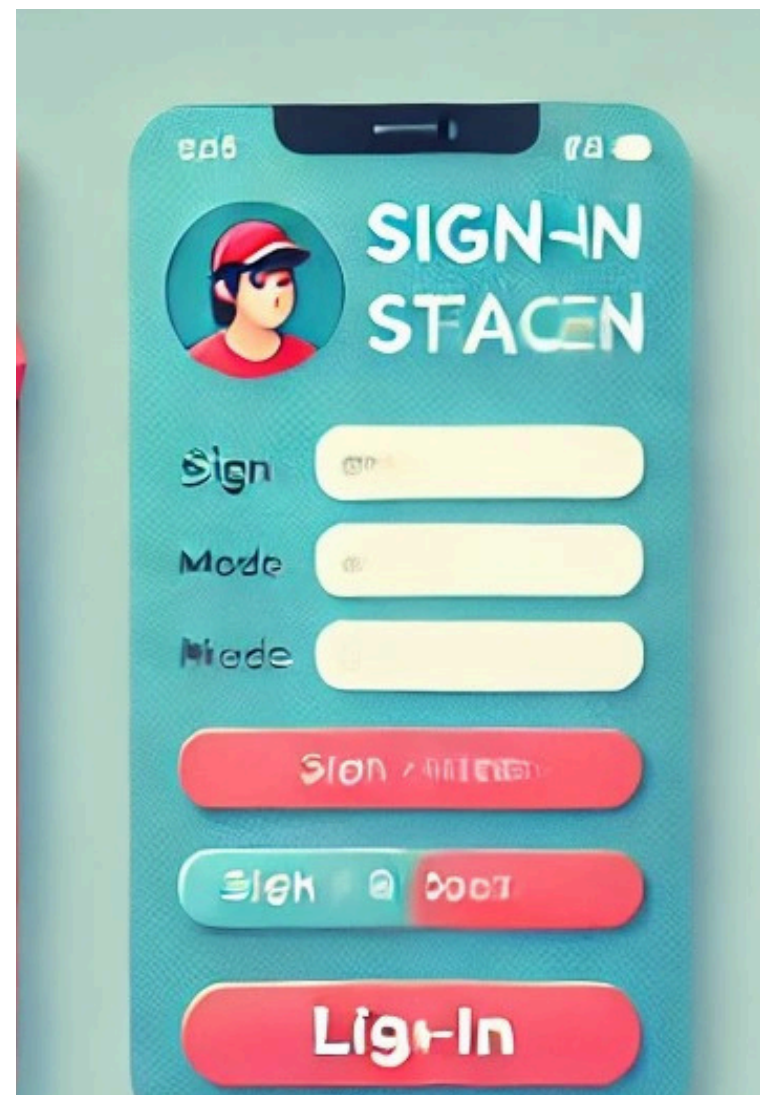
    return order;
```

CHAPTER 5 RESULTS AND DISCUSSION

LOGIN PAGE:



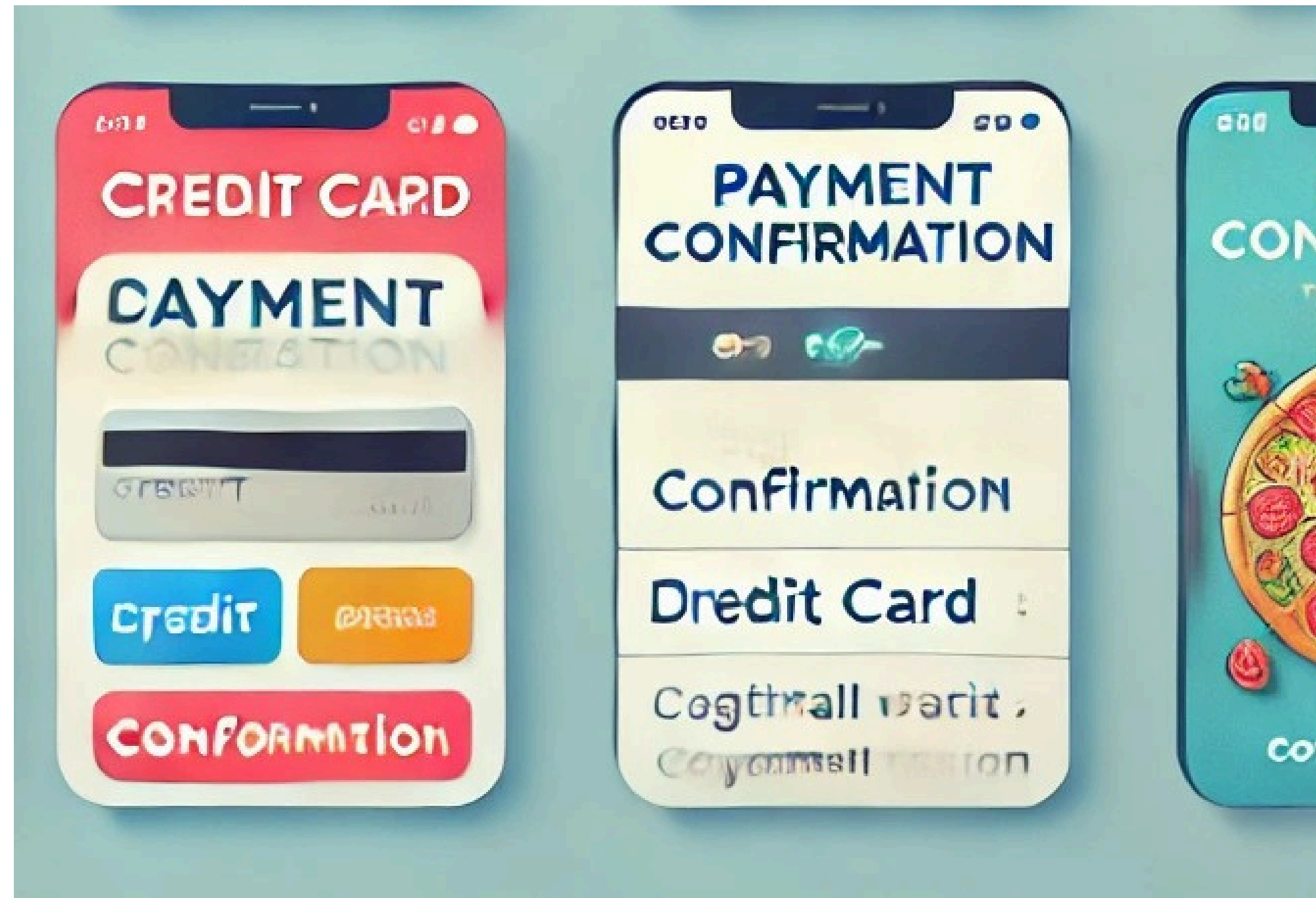
SIGNUP PAGE:



SELECTION PAGE :



PAYMENT PAGE:



CUSTOMER DETAILS PAGE



CHAPTER 6 CONCLUSION

Food delivery makes our life easier and more convenient. Delivery from restaurants to customers is also available. Delivery from small startup business like petty business is available and at the same time we can order packed food and fresh veggies

CHAPTER 7

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